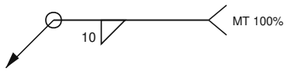
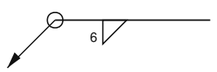
## TYPICAL WELD SYMBOLS (AWS A2.4) - arrow side below reference line

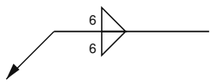
Drop drawbar box to rails/sill



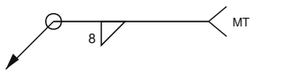

Pocket plates PL20 to box



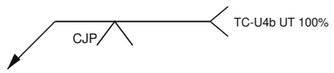

Crossmembers to main rails



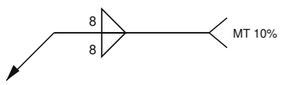

Bolster box seams (both sides)



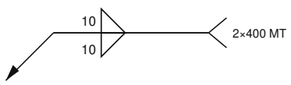

Centre plate to bolster



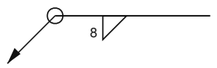

Horn cheek to chord (powered truck)



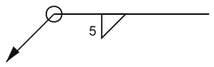

Transom box seams



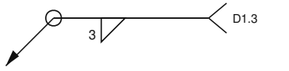

Motor bracket PL25 to transom



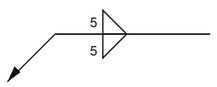

Engine / genset pedestals to tender rails




Coach crossmembers / outriggers




Window posts, roof bows HSS3x2x1/8




Stair stringers / treads (C-304)

Fillet sizes = leg (mm). ○ = weld all around. Tail = inspection. E70XX, prequalified WPS (D1.1 Cl. 5); preheat per Table 5.8 for A572-50 > 20 mm; grind toes on cyclic truck welds (cat. C→B').

**L4.2: safety-chain lugs PL44.5 (1-3/4 in) A572-50 faced to 42.5 mm at the jaw, 1-1/8 in Crosby G-2130 shackles + S-1325A; CJP (TC-U4b) + PL16 diaphragm; UT 100 %; preheat Table 5.8.**

## NOTES

1. Running-gear and drive fasteners: re-torque after first 10 h running; witness-mark all.
2. Structural slip-critical joints (if any added): ASTM F3125 A325, RCSC 2020 turn-of-nut.
3. CWI inspection of primary members (drawgear, trucks, drive brackets). Welders qualified to D1.1 / D1.3.
4. Zinc-plated fasteners: use dry K 0.20 column; lubricated (MoS2) K 0.14 column only where specified.
5. L4.2 P-103-1 motor hanger: L-weld 140 + 150, 10 mm fillet both sides + 2 PL16 gussets (6 mm both sides, structural); MT 100 % (Loco Bot L-103 fatigue check).

## FASTENER TORQUE TABLE (ISO 898-1, preload 0.75 × proof)

| Size | Class | As mm² | Preload kN | T dry K0.20 N·m | T lub K0.14 N·m |
|---|---|---|---|---|---|
| M10 | 8.8 | 58.0 | 25.2 | 50 | 35 |
| M10 | 10.9 | 58.0 | 36.1 | 72 | 51 |
| M12 | 8.8 | 84.3 | 36.7 | 88 | 62 |
| M12 | 10.9 | 84.3 | 52.5 | 126 | 88 |
| M16 | 8.8 | 157.0 | 68.3 | 219 | 153 |
| M16 | 10.9 | 157.0 | 97.7 | 313 | 219 |
| M20 | 8.8 | 245.0 | 110.2 | 441 | 309 |
| M20 | 10.9 | 245.0 | 152.5 | 610 | 427 |
| M24 | 8.8 | 353.0 | 158.8 | 762 | 534 |
| M24 | 10.9 | 353.0 | 219.7 | 1055 | 738 |

## APPLICATION SCHEDULE

| Joint | Bolt | N·m dry K0.20 | Locking |
|---|---|---|---|
| Axlebox keepers | M20 10.9 | 610 | Nord-Lock |
| Truck pivot cap | M24 10.9 | 1,055 | witness mark |
| Motor slide bolts | M20 10.9 | 610 | Nord-Lock; after jack-screw set |
| Brake caliper mounts | M20 10.9 | 610 | Loctite 243 |
| 45T split ring to hub flange (fitted DIN 609) | M16 8.8 | 219 | Nord-Lock; re-torque 10 h |
| MS05 mounting flange | 10.9 | per Poclain | AWAIT |
| Engine / genset isolators | M16 8.8 | 219 | prevailing-torque nut |
| Ballast pocket rods (P-106) | M20 8.8 | 120 | Nord-Lock |
| Clamp-bar set screws 106-CB1 | M16 | 4.8 | jam nut 60 + Loctite 243 |
| Seat anchors | M12 8.8 | 88 | through-bolt + backing PL |
| Stair pivot posts | M16 8.8 | 219 | Nord-Lock |
| Axle end caps | M16 10.9 | 313 | tab washer |
| Cab to frame | M16 8.8 | 219 | Nord-Lock |

**PE STAMP / SIGNATURE**

(blank - for reviewing NV PE)

**UNLESS NOTED: ISO 2768-mK general tolerances**
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

**PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)**

**Weld and fastener schedule (AWS A2.4 symbols, torques)**

Sheet S-503 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

**PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED**